Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 2 Quiz 1 - Kinematics in Two Dimensions

Mock Exam C · Form 2: Connecting ideas

Multiple Choice Questions: Select the best answer. (1 point each)

1. Two balls are launched simultaneously from the same height horizontally. B has 8 times A's horizontal velocity. Both land on the same ground without drag. Compare flight times and ranges.

(A) Both flight times and ranges are equal.

(B) Both B's flight time and range are 8 times A's.

(C) Flight times are equal; B's range is 8 times A's.

(D) B takes 8 times as long; ranges are equal.

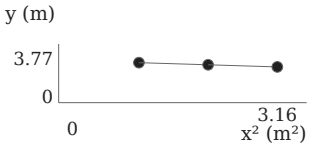
2. A ball leaves a 4 m ledge horizontally at 5 m/s. Use g=10 m/s² and consider airborne frames only. x² (m²) -> y (m): (1.16, 3.768); (2.16, 3.568); (3.16, 3.368). Which horizontal setting produces a reading of 3.668?

(A) 0.996

(B) 1.66

(C) 2.324

(D) 2.988



3. A student resolves a 14 m/s velocity directed 30° north of east. East is +x and north is +y. What is its x component?

(A) 24.249 m/s

(B) 6.062 m/s

(C) 12.124 m/s

(D) -12.124 m/s

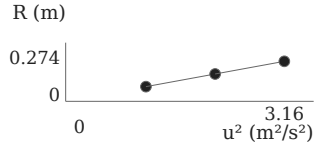
4. Launch at 30° and land at launch height, without drag. Use g=10 m/s². u² (m²/s²) -> R (m): (1.16, 0.1005); (2.16, 0.1871); (3.16, 0.2737). Set the horizontal origin at the first measurement. What is the new vertical intercept?

(A) 0.06028

(B) 0.1406

(C) 0.1005

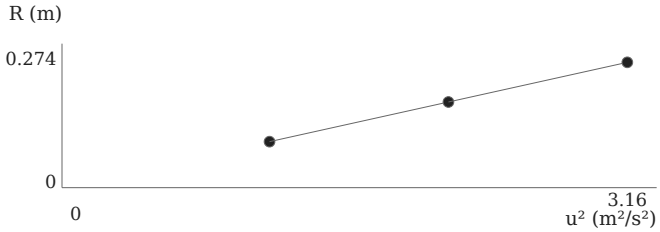
(D) 0.1808



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. Launch at 30° and land at launch height, without drag. Use g=10 m/s². u² (m²/s²) -> R (m): (1.16, 0.1005); (2.16, 0.1871); (3.16, 0.2737).

a. For the level-ground range setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Exchange the axes. Relate the new slope and its units to the original slope. **(3 pt)**

6. A projectile moves horizontally at +6 m/s while a target travels at +2 m/s. Track their horizontal separation during flight. t (s) -> xt-xp (m): (0.58, 9.68); (1.08, 7.68); (1.58, 5.68).

a. Build the model for moving target from physics and predict both slope and intercept. **(2 pt)**

b. Exchange the axes. Relate the new slope and its units to the original slope. **(2 pt)**

7. A ball leaves a 4 m ledge horizontally at 5 m/s. Use g=10 m/s² and consider airborne frames only. x² (m²) -> y (m): (1.16, 3.768); (2.16, 3.568); (3.16, 3.368). Build the model for airborne video frames from physics and predict both slope and intercept. **(3 pt)**